Rahul Yadav

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EXPERIENCE

Oracle Aug 2024 – Present
Senior Data Scientist (AI/ML Engineering) | Construction Intelligence Team Remote, PA, USA

Multi-Agent Platform for Fragmented Construction Project Data (internal proof of concept)

- Architected multi-agent system unifying **5** data modalities (schedule, cost, contract/BIM, video, audio) to explain predictions
- Built orchestrator on OpenAI Agents SDK coordinating **7** specialist sub-agents over a shared Apache AGE knowledge graph
- Built RAG agent on Oracle 23ai DB across contracts, RFIs, and BIM with hybrid BM25 + vector search at **<200ms** P95 latency
- Built Kafka consumers normalising vision events into the knowledge graph, correlated against P6 activity baseline
- Established LangSmith LLM-as-judge eval harness gating CI/CD across all **8** agents on quality, latency, and cost

Oracle Construction and Engineering Advisor for Safety: AI Safety Risk Forecasting

- Lead a team of **4** data scientists and own modeling decisions for Oracle's AI safety forecasting product (GA Mar 2026)
- Framed the ML problem as weekly incident-risk forecasting that ranks the portfolio projects most likely to have an incident
- Directed transformation and analysis of multi-source project safety data into consistent, model-ready weekly indicators
- Built industry model on construction project data, enabling forecasts from day one of implementation

Hilabs Feb 2024 – May 2024
Senior Data Scientist | Applied Data Science Team Washington D.C., USA

LLM-Powered Contract Intelligence for Healthcare Claim Adjudication

- Delivered document understanding system extracting parties, terms, and pricing rules across **5**+ insurance contract types
- Achieved **87%** F1 on entity extraction by building custom LayoutLM pipeline for joint text/table/image processing
- Deployed self-hosted fine-tuned Mistral on private VPC and retrieval on OpenSearch, meeting HIPAA and PHI requirements
- Delivered V1 to **3** enterprise insurance clients, validating pricing accuracy and fit with their claim processing applications

Apexanalytix May 2023 – Feb 2024
Data Scientist (Intern → FTE) | Analytics Team Remote, New York, USA

Enterprise RAG Assistant for Internal & Client Knowledge Access

- Achieved **89%** human-eval approval by building production RAG chatbot on LangChain, ChromaDB, and OpenAI generation
- Lifted retrieval relevance by **37%** over flat-chunk baseline on internal eval set via parent-child chunking with MMR scoring
- Built feedback collection pipeline driving iterative tuning of prompts, retrieval, and reranking based on real user query patterns
- Drove adoption across **4 internal and 5 external** client applications, replacing manual document search across the org

Oracle Financial Services Software Sep 2020 – Aug 2022
Senior ML Engineer | Oracle Financial ML Team Bengaluru, India

Preemptive Anomaly Detection for Corporate Billing (ORMB) and ML-Driven Collections (OFSLL)

- Hit **92%** precision on 1.2M bills and 5M segments in **20 min** via in-memory multivariate detection on Oracle in-database ML
- Built semi-supervised classifier with local + global model explainability for analyst-facing audit and override workflows
- Shipped into Oracle ORMB, deployed across banking, insurance, and telco customers running the platform globally
- First-named inventor on US Patent **12,333,585** [US12333585B2], anomaly detection for enterprise billing
- Designed ML collections method behind US Patent **12,002,091**; shipped as Intelligent Segmentation in OFSLL

Wipro Limited Jun 2018 – Sep 2020
AI/ML Engineer | AI Research Team Bengaluru, India

NLP Ticketing Assistant for Employee Support Operations

- Cut human-agent intervention by **74%** and wait time from **18** → **4 min** via BERT-powered semantic search over ticket corpus
- Hit **79%** accuracy across **10+** ticket categories via intent classifier and resolution retrieval, powering auto-routing in production
- First-named inventor, US Patent **11,386,712** (multimodal emotion recognition); corresponding author, Springer ICMLIP 2019

EDUCATION

University at Buffalo, SUNY Aug. 2022 – Dec. 2023
Master of Science, Computer Science and Engineering – CGPA 3.82 / 4.0 Buffalo, NY, USA

Indian Institute of Technology (BHU), Varanasi Jul. 2014 – May 2018
Bachelor of Technology, Electronics Engineering Varanasi, India

PATENTS & PUBLICATIONS

6 Granted · 3 Pending · 1 Paper

Inventor on 9 USPTO filings spanning anomaly detection, NLP topic categorization, document AI, data quality, and multimodal emotion recognition; first-named inventor on three granted patents; five of six granted patents from Oracle Financial Services (2020–2022).

- **Granted:** US 11,386,712 - Multimodal Analysis Based Emotion Recognition [US11386712B2]
- **Granted:** US 12,002,091 - Technology System for Assisting Financial Institutions in Debt Collection [US12002091B2]
- **Granted:** US 12,333,585 - Anomaly Detection for Bill Generation (productized in Oracle ORMB) [US12333585B2]
- **Granted:** US 12,443,792 - Reference Driven NLP-Based Topic Categorization [US12443792B2]
- **Granted:** US 12,340,613 - Framework for Identifying Documents [US12340613B2]
- **Granted:** US 12,694,306 - ML Pipeline with Data Quality Framework (granted 28 Jul 2026) [US12694306B2]
- **Pending:** 18/056,456 - Predicting Record Topic Using Transitive Relations [US20240169216A1]; 17/989,202 - Hierarchical Visualization of Clustered Datasets [US20240168979A1]; 19/320,494 - continuation of US 12,443,792 [US20260004063A1]
- **Publication:** Virtual Conversation with Real-Time Body Movement / Gesture Prediction - *ICMLIP 2019* [link]

PEER REVIEW & JUDGING

4 Completed Reviews · 3 Venues · 2026

- **TMLR** (Transactions on Machine Learning Research): invited by an Action Editor, May 2026; **2** full reviews completed
- **NeurIPS 2026**: invited July 2026 after the TMLR reviews; **1** full review completed, acknowledged by the Program Chairs
- **IEEE Access**: invited directly by the Associate Editor, Oct 2026; review completed 23 Oct 2026, verified on Web of Science
- **Oracle AI Review board** (internal): nominated Nov 2024; review AI models from other Oracle business units pre-production

PROJECTS

A2A + MCP Multi-Agent Workflow [code] 2025 – 2026

- Multi-agent system using Agent-to-Agent (A2A) communication and Model Context Protocol (MCP) for tool sharing
- Structured agent handoffs for task decomposition, parallel execution and result synthesis as a production A2A reference

Dynamic Agent Spawning Framework [code] 2026

- Runtime agent creation system where LLM dynamically decides agent type, tool set, and prompts per request, no static roster
- Scales agent count and complexity to task difficulty, cutting idle compute vs fixed multi-agent fleets on lightweight queries

From RAG POC to Production: What Actually Changes [blog] Technical Blog · 2026

- Lessons from shipping production RAG: hybrid retrieval beats vector-only, hallucinations are a product problem, observability before launch, tiered model selection (50% cost cut) and semantic caching (30–50% hit rate)

SKILLS

Agentic & LLM Systems : OpenAI Agents SDK, LangGraph, LangChain, LlamaIndex, MCP, A2A, RAG, GraphRAG, Tool-use Agents, LLM-as-Judge, LoRA / QLoRA Fine-tuning, LangSmith, Evaluation Harnesses

ML, NLP & Vision : PyTorch, TensorFlow, Hugging Face, scikit-learn, Whisper, LayoutLM, Sentence Transformers, Embeddings & Re-rankers, Quantization & Distillation, MLflow, YOLOv11, OpenCV, CLIP

Data, Cloud & MLOps : Kafka, PySpark, Snowflake, SQL, Oracle 23ai Vector DB, Elasticsearch, Apache AGE, Neo4j, OCI, AWS, Azure, GCP, Kubernetes, Docker, Terraform, FastAPI, CI/CD

Programming Languages : Python, SQL, JavaScript, Java